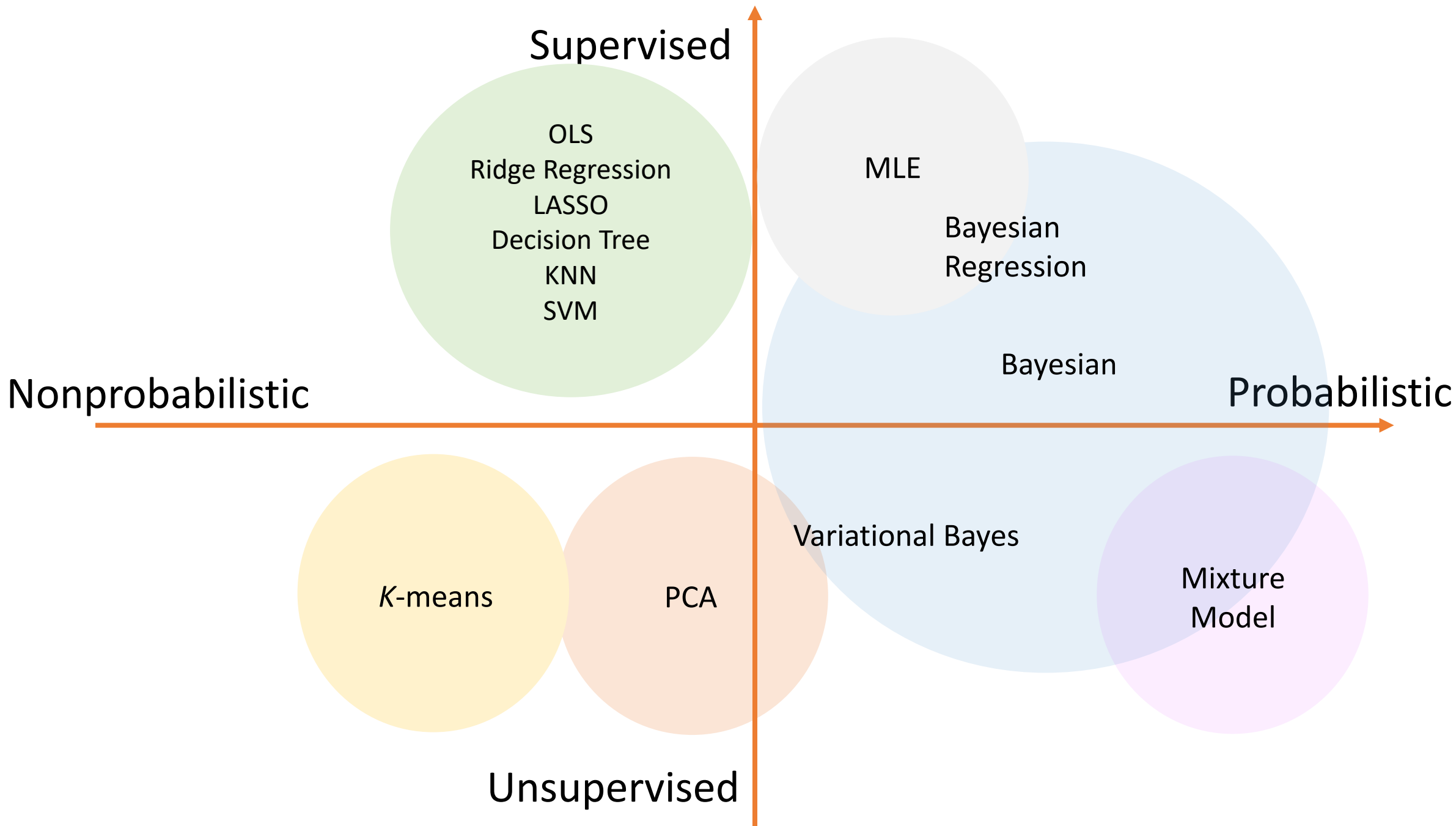


A First Course in Machine Learning

Chapter 7

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Two Unsupervised Classes

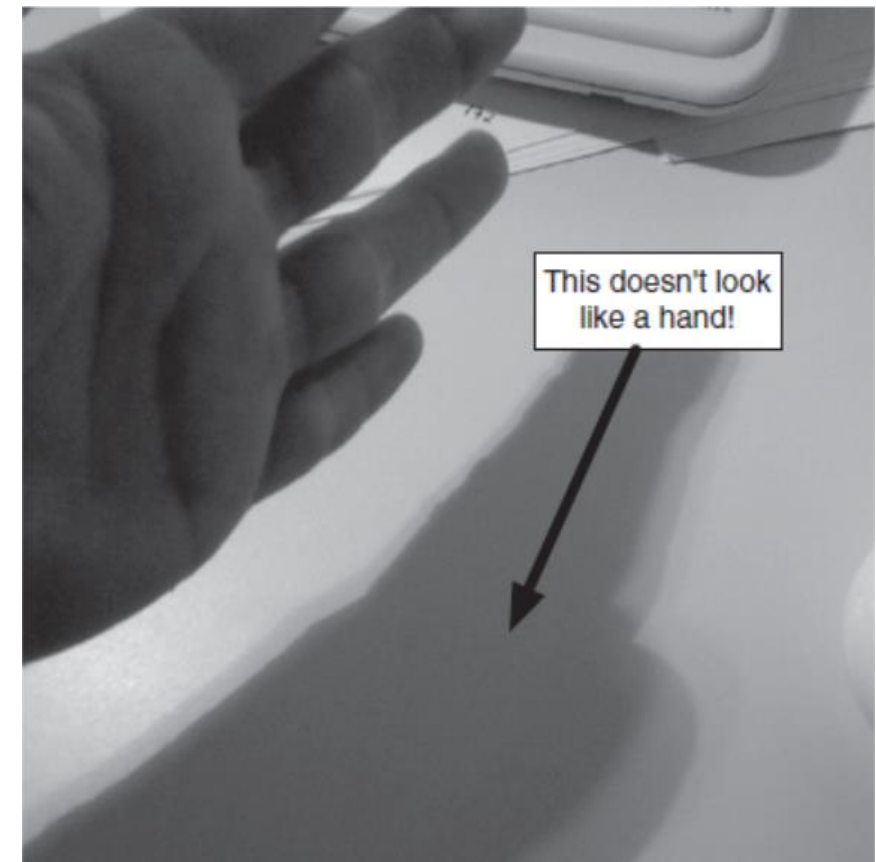
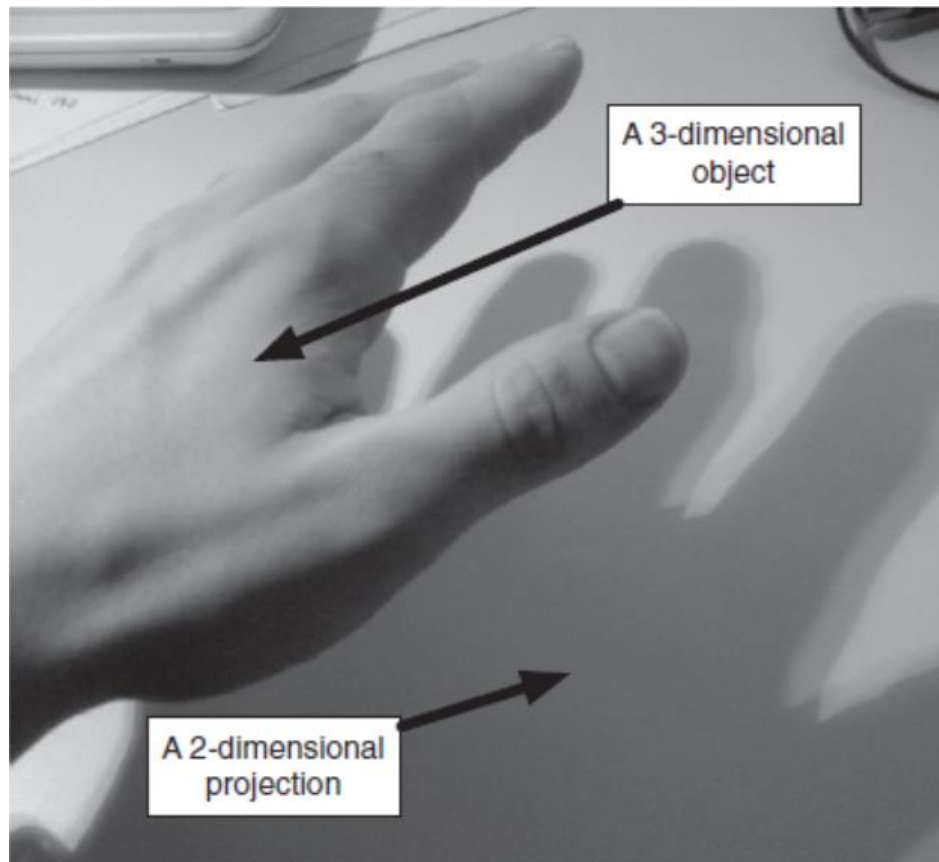
- In the previous chapter we introduced clustering methods. Now we turn to a second class of unsupervised methods: projection techniques.
- We will take datasets in very high dimensions and project them down to a smaller number of dimensions for visualization and feature selection.
- These techniques fall within the scope of *latent variable models*.

The General Problem

- We start with a dataset of N objects, where each object $\mathbf{y}_n$ is an M -dimensional vector.
- The number of parameters increases with the number of dimensions, M . Therefore, if M is large, it is challenging to make parameter estimation and data visualization.
- Often useful to transform the M -dimensional $\mathbf{y}_n$ into a D -dimensional $\mathbf{x}_n$ ($D < M$), known as *projection*.

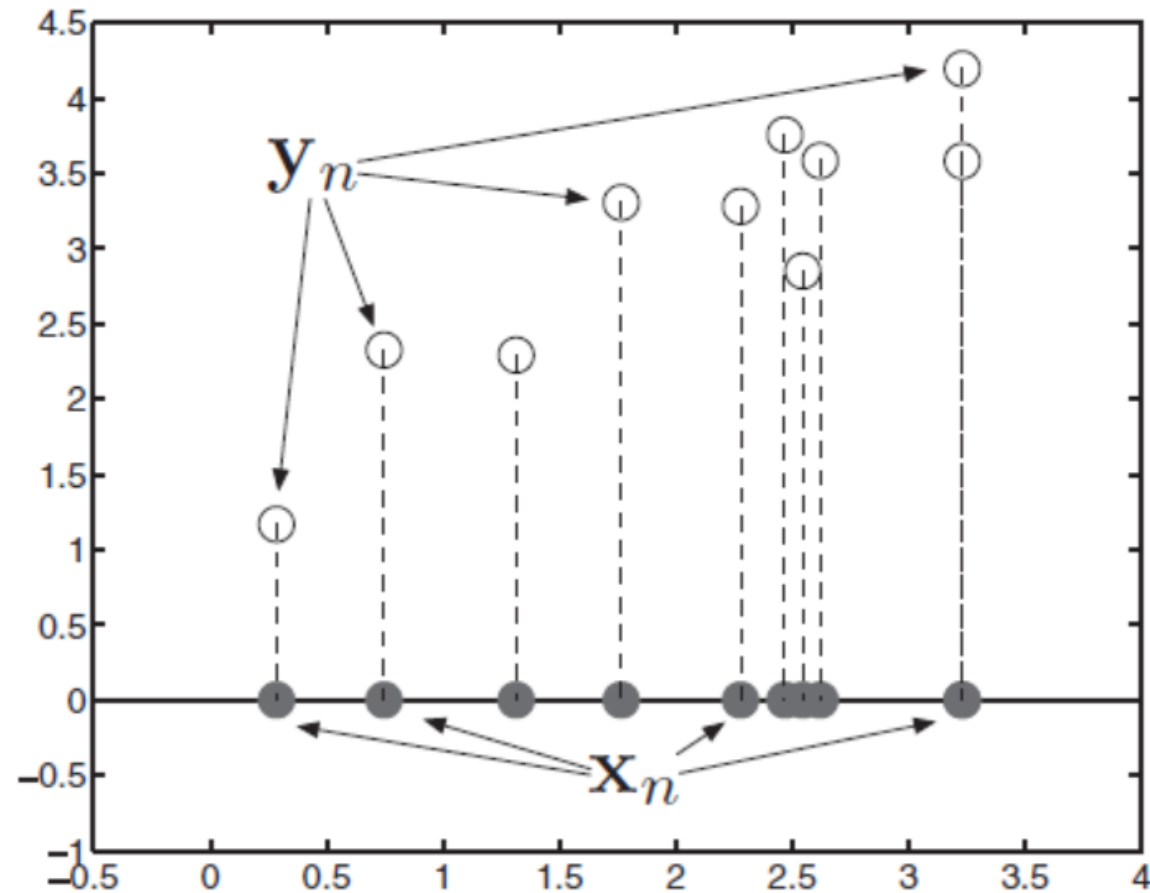
Dimensionality Reduction

- We would like to retain as much of the interesting structure in our data as possible when performing the projection.



Mathematical Projection

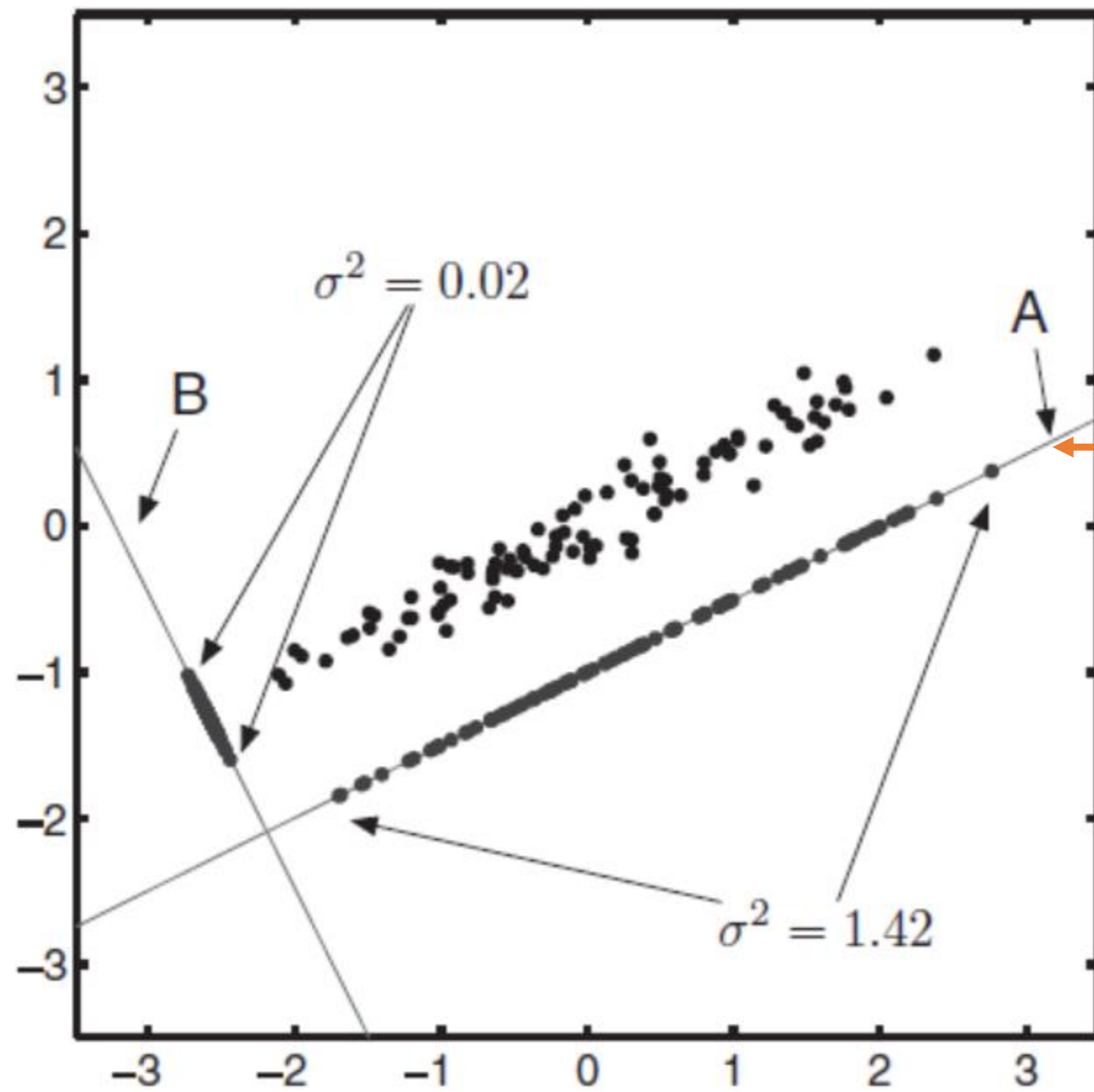
- The projected dimension happens to be one of the original two dimensions, but this is not necessary.



Variance of Projections

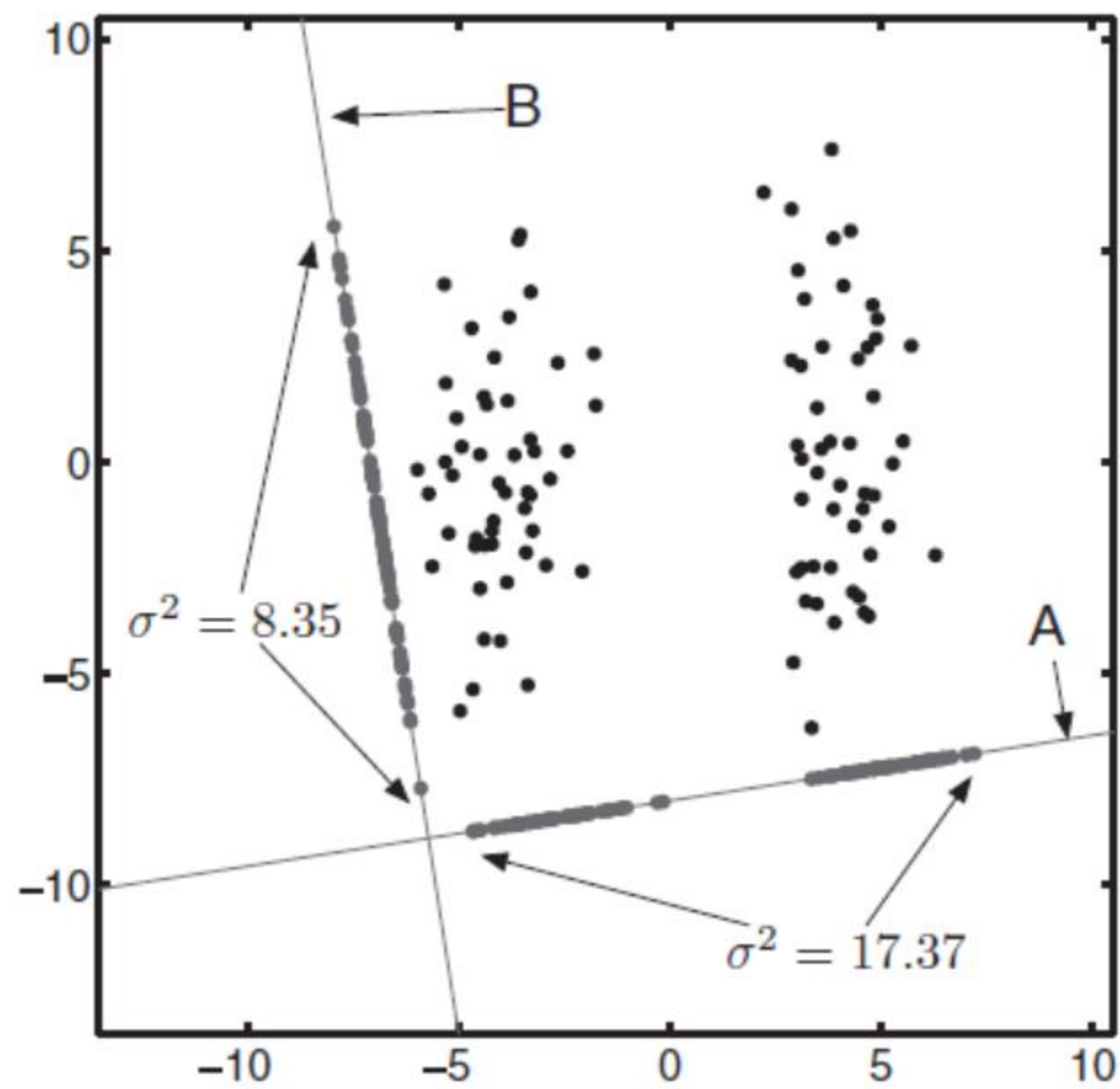
- Q: What do we mean by interesting structure?
- When data has been projected into two lines, A and B , each of which is given by a linear combination of the original dimensions: $\mathbf{x}_n = w_1 y_{n1} + w_2 y_{n2} = \mathbf{w}^T \mathbf{y}_n$.
- The variance of the data in each one-dimensional space is

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_x)^2$$



Choosing Projection

- In Figure 7.2(a), it is obvious that the variance of the data after projection A will be higher than that for projection B .
- Given no other information, if we must project into one-dimension, we should probably choose A .
- Or, if we are forced to throw one of A and B away, it feels safer to throw away the information present in B .

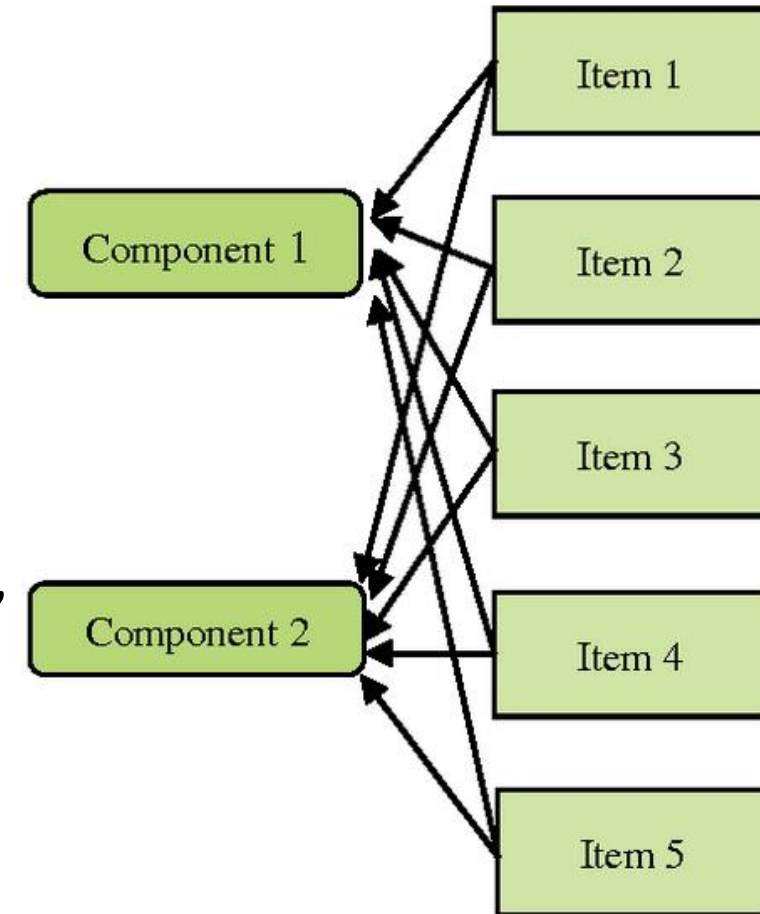


Clustering Data

- In Figure 7.2(b), projecting onto A preserves cluster structure of data whilst projecting onto B does not.
- The variance of the data after projection A is more than double that for projection B due to the cluster structure—all points are farther from the mean.
- Therefore, the *variance after projection* is the quantity to be maximized when deciding on projection directions in Principal Components Analysis (PCA).  Similar but not the same as Factor Analysis

Principal Components Analysis

- PCA is a data reduction technique—to capture the variance in variables in a smaller set.
- The whole point of the PCA is to create one or more *index variables* using a linear combination in an optimal way.
- Get the optimal number of components, the optimal choice of measured variables for each component, and the optimal weights.



Principal Components Analysis

- Principal components analysis (PCA) is perhaps the most widely used statistical technique for projecting data into a lower-dimensional space.
- It is very popular within machine learning for (i) visualization and (ii) feature selection.
- PCA defines a linear projection: each of the projected dimensions is a linear combination of the original dimensions.

Learning Task

- When we are projecting from M to D dimensions, PCA will define D vectors, $\mathbf{w}_d$, each of which is M -dimensional.

- The d th element of the projection is computed as

$$x_{nd} = \mathbf{w}_d^T \mathbf{y}_n.$$

- The learning task is to choose *how many dimensions we want to project into (D)* and then *pick a projection vector ($\mathbf{w}_d$) for each*.

Maximal Variance

- PCA uses variance in the projected space as the criterion to choose $\mathbf{w}_d$.
- $\mathbf{w}_1$ will be the projection that makes the variance in the x_{n1} as high as possible. The second projected dimension is also chosen to maximize the variance but $\mathbf{w}_2$ must be orthogonal to $\mathbf{w}_1$ ($\mathbf{w}_1^T \mathbf{w}_2 = 0$).
- $\mathbf{w}_3$ must maximize the variance and be orthogonal to both $\mathbf{w}_1$ and $\mathbf{w}_2$. In general,

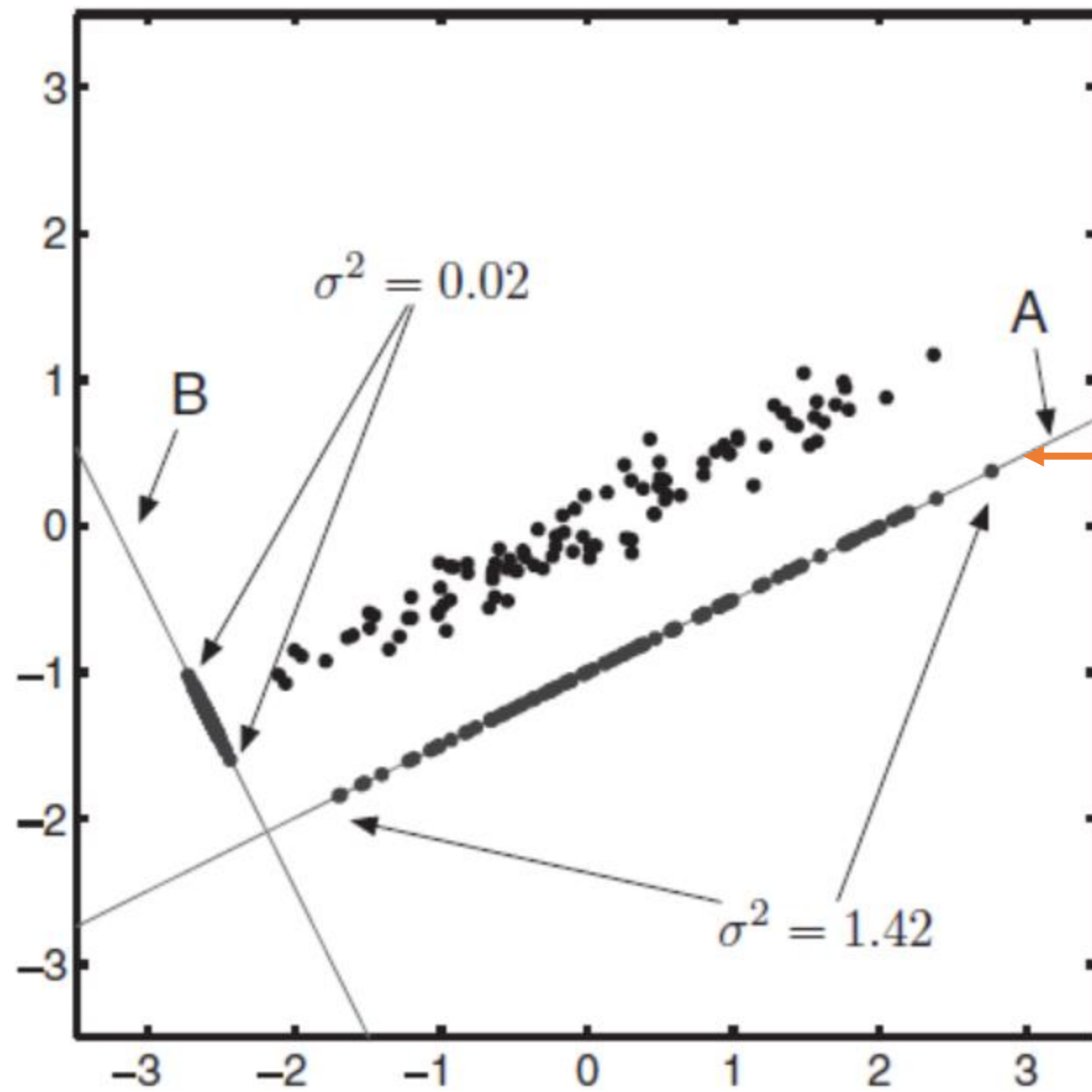
$$\mathbf{w}_i^T \mathbf{w}_j = 0, \forall j \neq i.$$

Mean Centering

- This set of constraints tells us that, if we set $D = M$, performing PCA amounts to rotating a rotation of the original data, without any loss of information.
- Since it is only the direction of each $\mathbf{w}$ that is important, PCA imposes the constraint that each $\mathbf{w}_i$ must have a length of 1, $\mathbf{w}_i^T \mathbf{w}_i = 1$.
- It is useful to subtract the mean from each $\mathbf{y}_n$ so that each of the original dimensions has zero mean, i.e., $\bar{\mathbf{y}} = \sum_{n=1}^N \mathbf{y}_n / N = 0$.

Summary of PCA

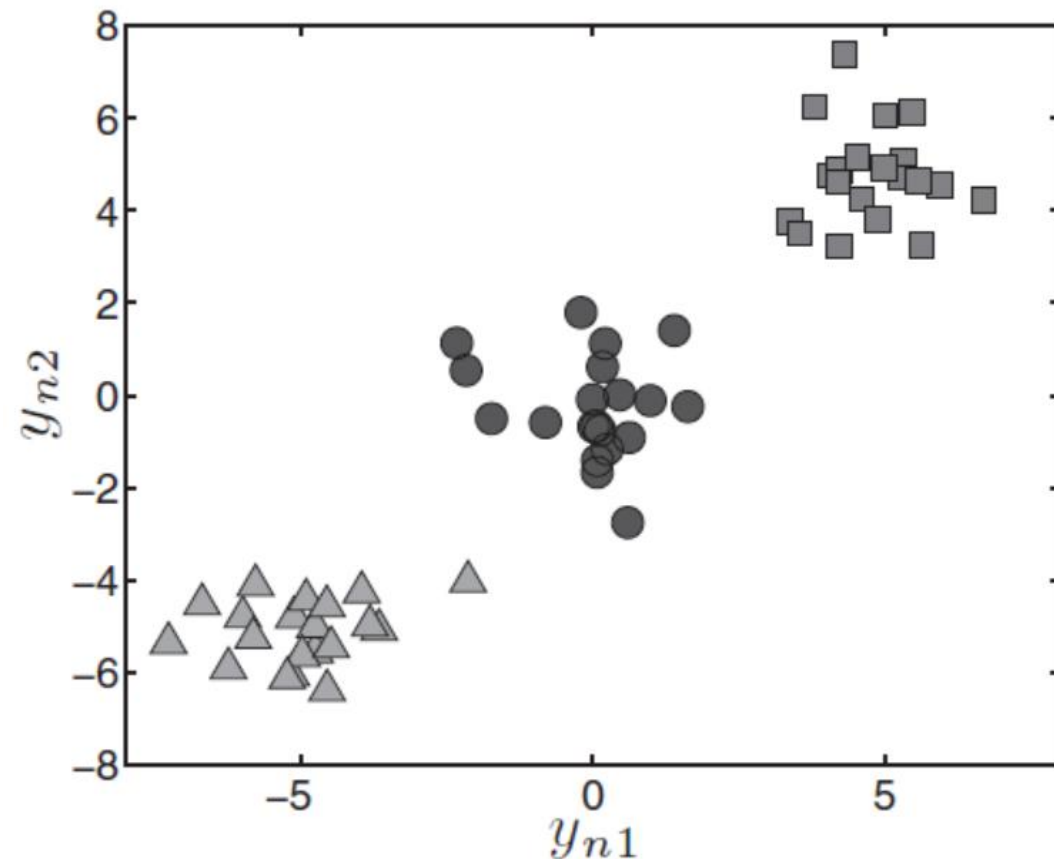
- Performing PCA on a set of data points requires the following steps:
 1. Centering each object
 2. Compute the sample covariance matrix $\mathbf{C} = \frac{1}{N} \mathbf{Y}^T \mathbf{Y}$
 3. Find the M eigenvector/eigenvalue pairs of $\mathbf{C}$, which can be done using `np.linalg.eig()`.
 4. Find the eigenvectors corresponding to the D highest eigenvalues, $\mathbf{w}_1, \dots, \mathbf{w}_D$.
 5. Create the d th dimension for objects in the projection, $x_{nd} = \mathbf{w}_d^T \mathbf{y}_n$ (or $\mathbf{X} = \mathbf{Y}\mathbf{W}$ where $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_D]$ and $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T$).



Can you find errors with these two principal components?

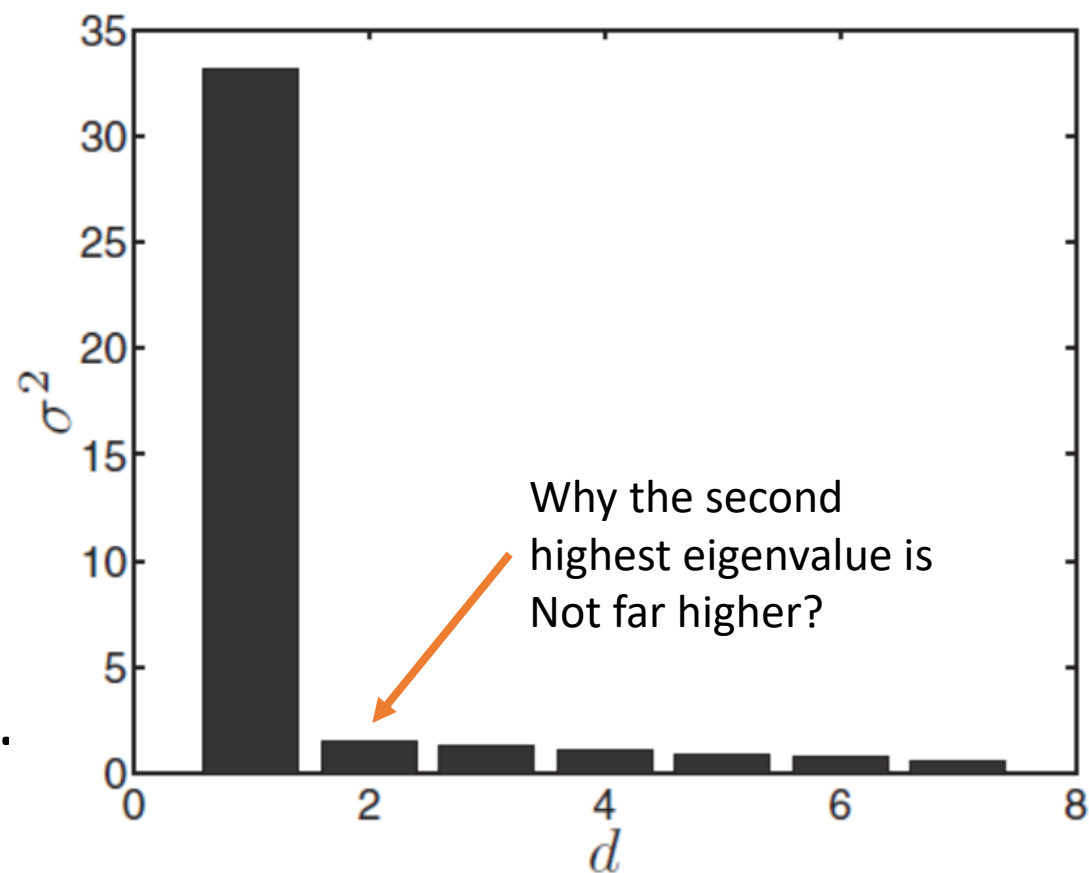
Example 1

- We add additional five dimensions, $y_{nd} \sim \mathcal{N}(0,1)$, $d = 3, \dots, 7, n = 1, \dots, N$, to a two-dimensional dataset with three clusters.
- That is, there is structure in the first two dimensions and noise in the remainder.
- In reality, we don't know which features are of our interest.



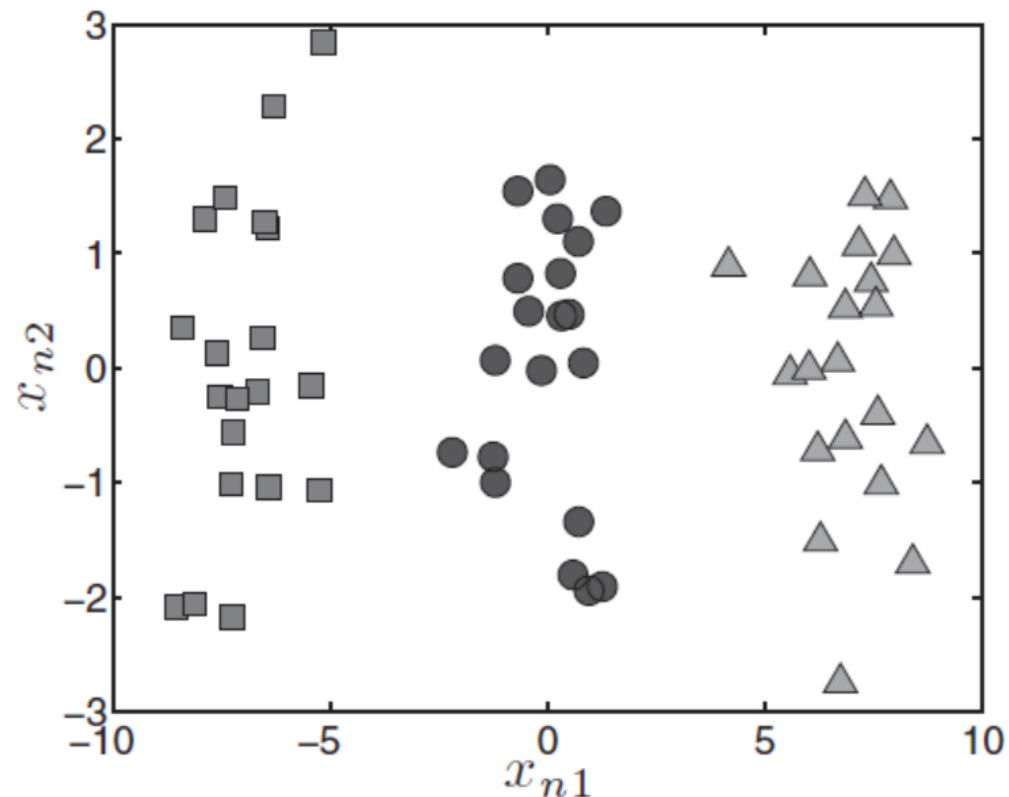
Unique High Eigenvalue

- After centering the data, we can have the values of seven eigenvalues of the covariance matrix, $\mathbf{C} = \mathbf{Y}^T \mathbf{Y}$, ordered by magnitude.
- Recall that these values correspond to the variance in each of the D potential projection dimensions.
- The highest eigenvalue is far higher than any of the others.



Projected Data

- The eigen-spectrum (magnitudes of the eigenvalues) tells us that one projected dimension will suffice..
- This is because the clusters all lie on the line $y_{n1} = y_{n2}$.
- After plotting the data in the first two projected dimensions, we can see that the first projected dimension x_{n1} holds all the cluster structure.



PCA is Unsupervised

- Notice that we did not use the label information of data objects, though we plotted them with different symbols (circles, squares, etc.).
- Since we will not always get plots that send as clear a message as above examples, the eigen-spectrum provides some useful information but its interpretation is highly *subjective*.
- In reality, D can be chosen based on the classification performance with $\mathbf{X}$ via cross-validation.